

Increasing Access to Electricity with DER-Powered Microgrids

Hayden Coke and Logan MacFarland

DATA 510 Data Science Capstone · Summer 2026

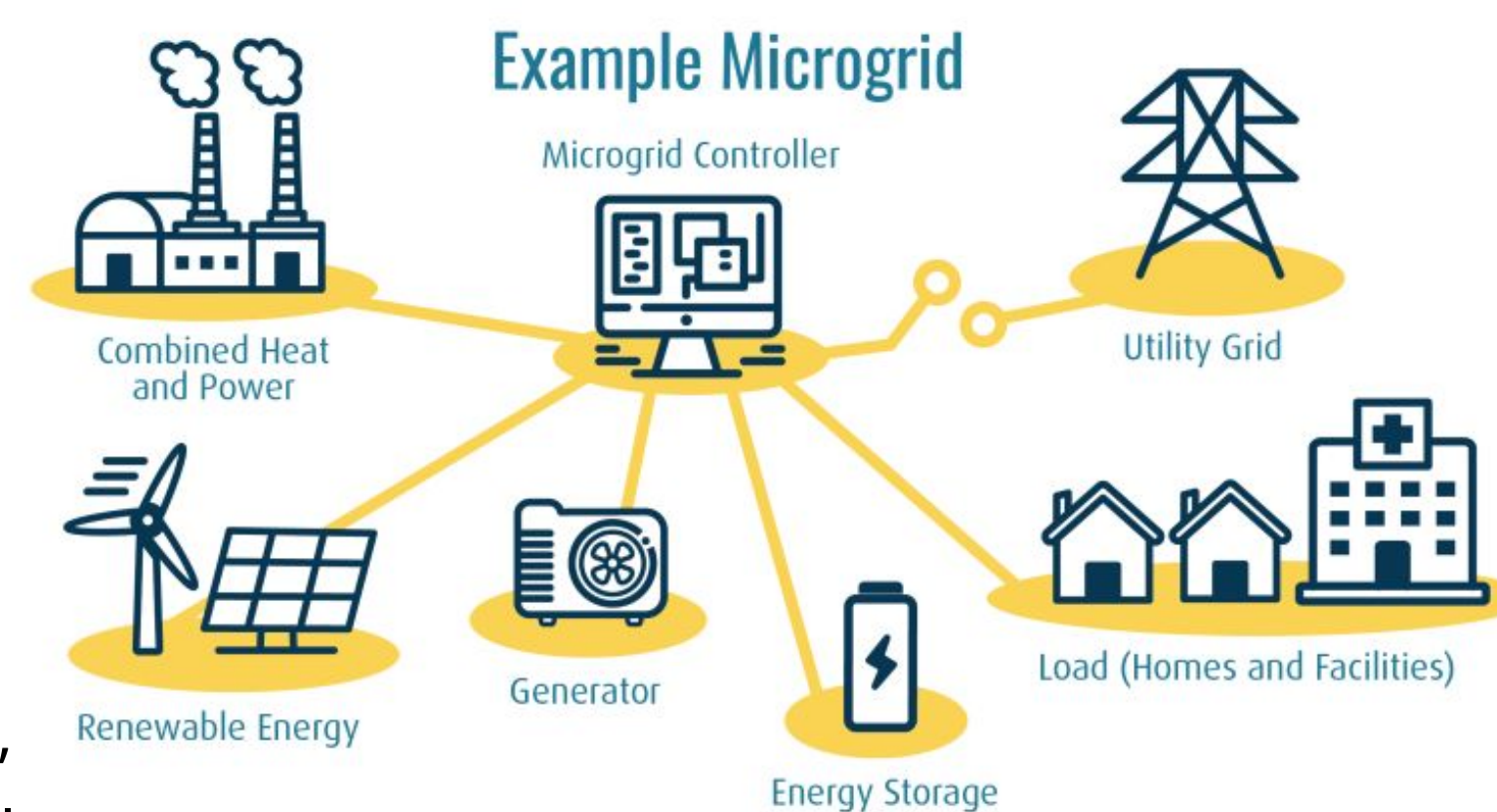
The Problem

When communities lack access to reliable power, their most vulnerable members may risk losing access to:

- Medical Care
- Heating and Cooling
- Cooking
- and other necessities

These interruptions can range from inconvenient to life threatening.

Using Distributed Energy Resources, communities in need of more reliable access to power can install a **microgrid** that can operate independently of the main grid in an emergency.



Which zip codes may benefit from the installation of microgrids powered by Distributed Energy Resources such as wind and solar?

With ever increasing reliance on electricity, the demand for energy is not always met by our current infrastructure. Microgrids powered by Distributed Energy Resources can provide power to entire communities in a more reliable and efficient manner than traditional grid connection. Knowing where microgrid installation is most feasible is a step toward increasing grid resiliency for communities in need.

Impact & stakeholders:

- Microgrids powered by Distributed Energy Resources can serve neighborhoods, hospitals, university campuses, and even small towns.
- If a community wishes to install a microgrid, they must collaborate with their local energy company to integrate the microgrid with the main power grid.

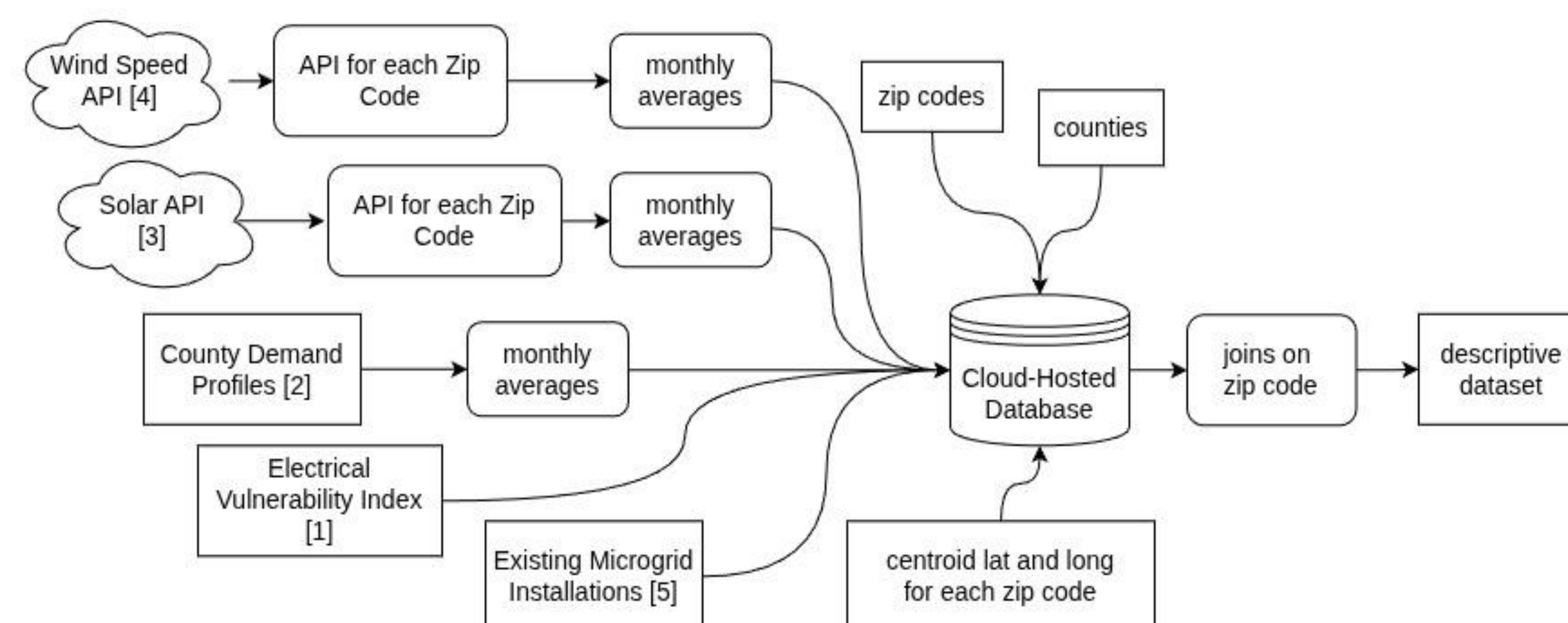
Ethics & responsible use:

Energy resiliency is a subject that can have a big impact on vulnerable communities, such as those who rely on access to electricity for medical needs or rural communities who are underserved by existing infrastructure.

Data Engineering

Sources:

- Mapping the electric vulnerability index: A metric for targeting energy storage resilience solutions [1]
- Hourly Electricity Demand Profiles for Each County in the Contiguous United States [2]
- National laboratory of the Rockies PVWatts API [3]
- Climate Data Store's ERA5 hourly wind data [4]
- Onsite Energy Installation Database [5]



Reproducibility & References

Repository: https://github.com/wu-msds-capstones/project-writeup-hayden_logan_m · Key dependencies: Python 3.12.3, pandas, scipy, R 4.6.1, ggplot, tidyverse

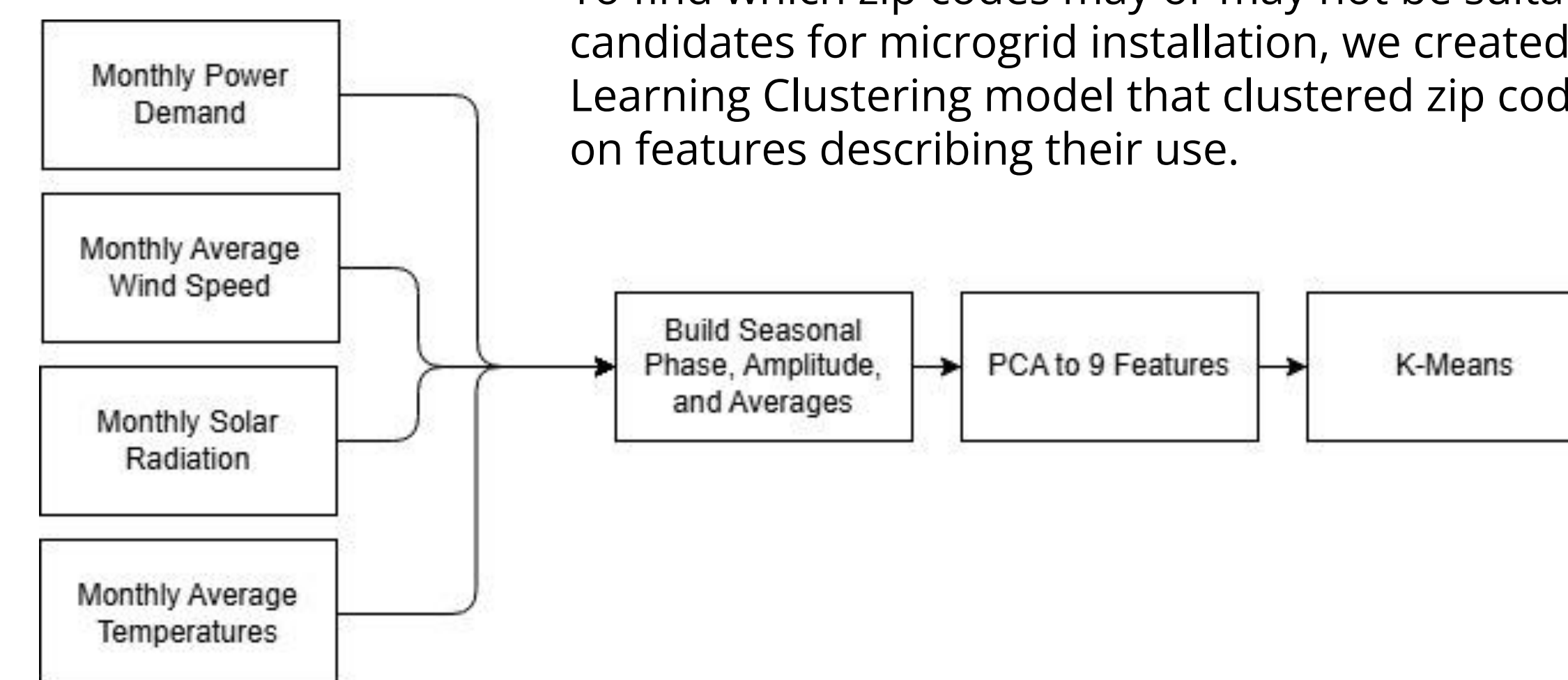
References: [1] Kerby et al., Mapping the electric vulnerability index: A metric for targeting energy storage resilience solutions, 2025. [2] Obika, Kodi, Wesley Cole, and Marie Rivers. Hourly Electricity Demand Profiles for Each County in the Contiguous United States. National Laboratory of the Rockies (NLR), Distributed by Open Energy Data Initiative (OEDI), 2025. [3] National laboratory of the Rockies, PVWatts API, U.S. Department of Energy. [4] ERA5 hourly data on single levels from 1940 to present, Climate Data Store. [5] Onsite Energy Installation Database, U.S. Department of Energy, ICF, Oak Ridge National Laboratory (ORNL)

Microgrid graphic: <https://www.smpa.com/microgrids-community-resilience-and-climate-action>

Background image: https://columbiacountyobserver.com/master_files/County_News_2018/19_0226_fpl-officially-unveils-columbia-county-solar-farm.html

Methods & Evaluation

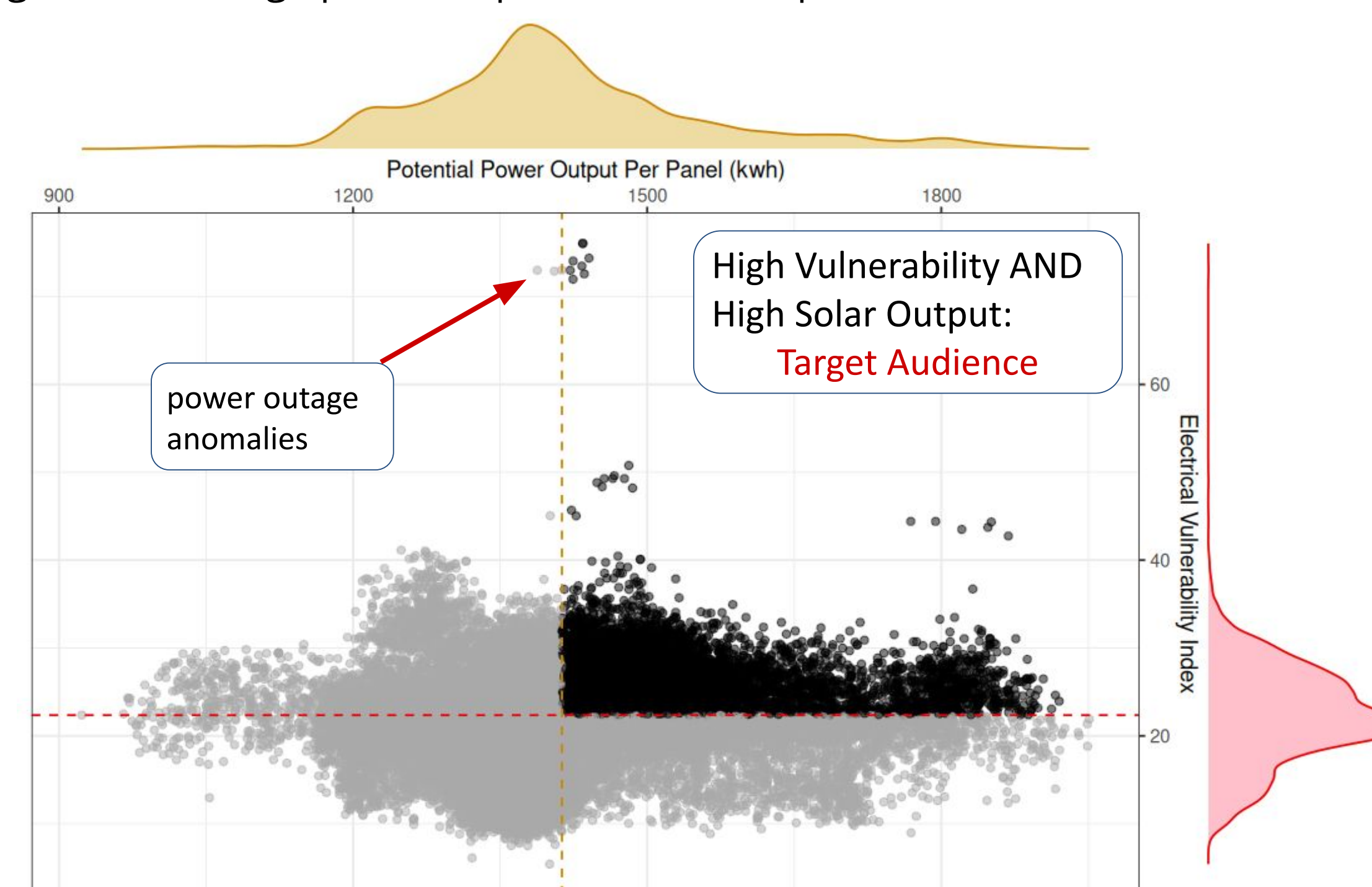
Approach:



To find which zip codes may or may not be suitable candidates for microgrid installation, we created a Machine Learning Clustering model that clustered zip codes based on features describing their use.

Validation:

- We used above-average EVI to determine which zip codes were in high need of microgrid installation. 14161 zip codes were identified as having above-average vulnerability.
- Out of the high need zip codes, 6946 of those zip codes were also identified as having above-average potential photovoltaic output.



Conclusions and Limitations

Conclusions:

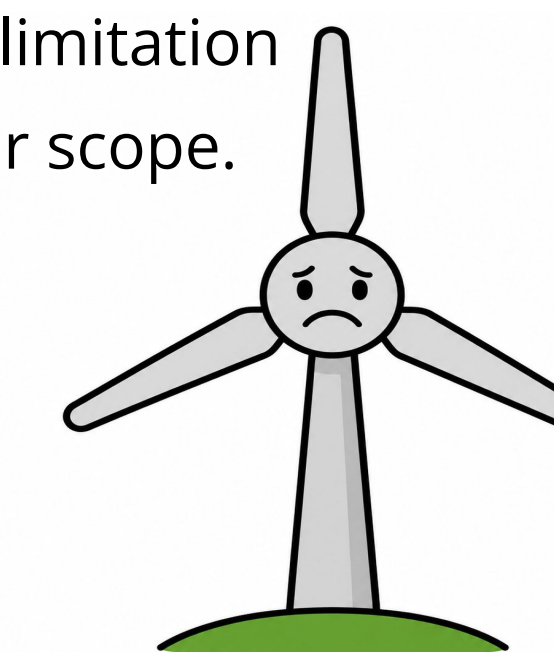
While we were able to see which regions had high solar potential, a major limitation of our analysis was that microgrids have various motivations outside of our scope.

- Some microgrids are simply battery installations
- Some microgrids have space limitations
- Our monthly, zip code wide analysis was not granular enough

Limitations:

Using monthly wind speed averages for entire zip codes was not an effective way to analyze potential for wind installations

- Wind power is proportionate to the wind velocity *cubed*. For two locations with the same average wind speed, a location with strong gusts will yield much more power.
- Average wind speeds are notably under-estimated in the ERA5 dataset
- Wind installations are usually prioritized in high gust areas in a region, like a natural wind corridor, which is unlikely to be observed at our zip centroid



Results & Visualizations

Cluster analysis was done on features of each zip code, including power demand by zip as well as solar and wind potential.

- Most clusters represent broad regional groupings
- High demand urban areas extract into their own clusters
- Microgrids in the same cluster offer similar resource access

Geographic Representation of Clusters

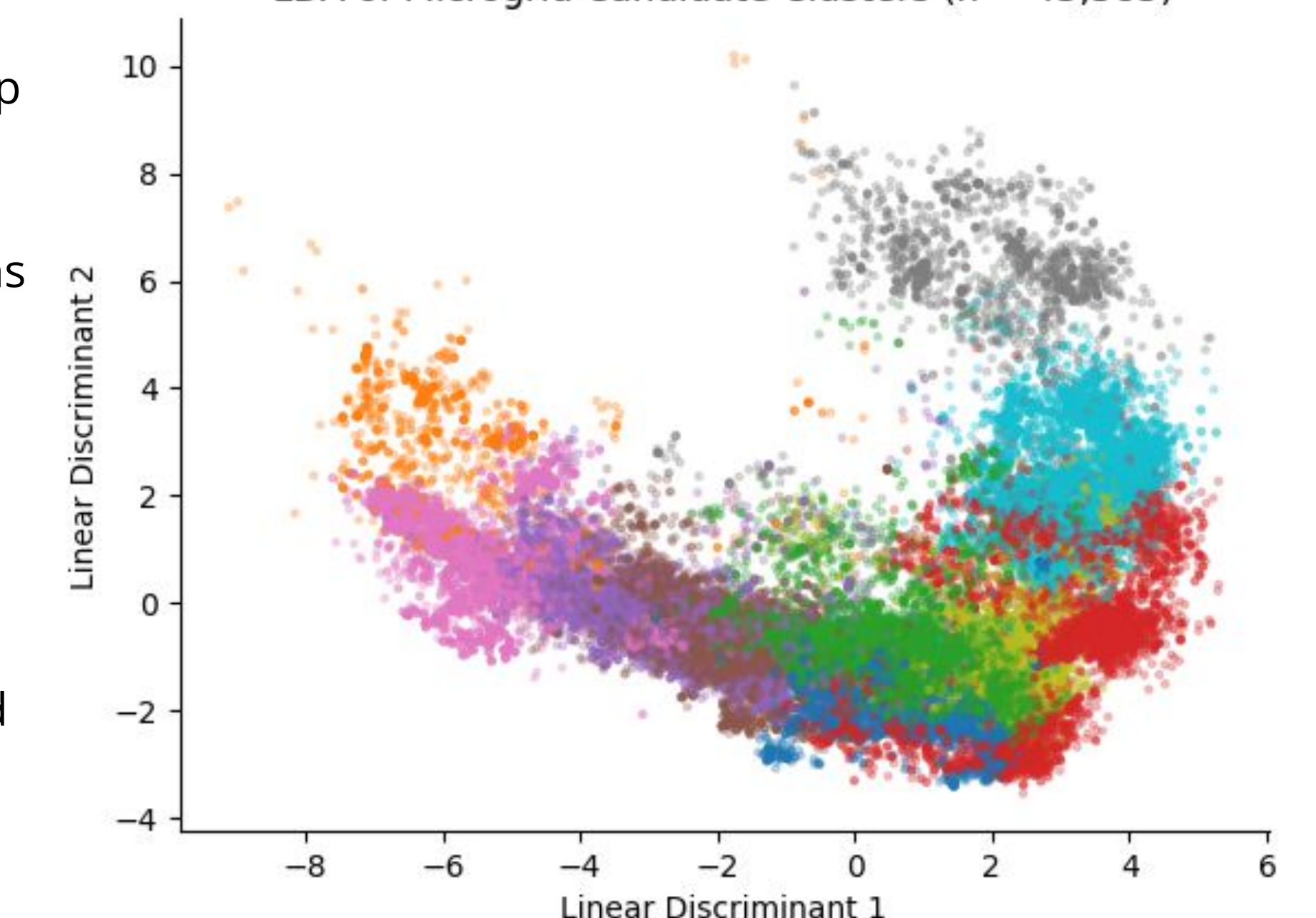


Geographic projection shows how most microgrid candidate locations are similar to neighbors

Linear Discriminant Analysis shows how the features of a zip code are represented in the data

- High demand, urban regions form their own regions
- Most clusters are relatively connected where the gradual changes in regional features
- Notably the green cluster is high demand, high solar locations where a microgrid would thrive

LDA of Microgrid Candidate Clusters (n = 43,585)



Future Work

Analysis of which microgrid installations can benefit from various renewable resources would start with a more granular analysis of both time series and geocoding.

- Wind analysis needs to consider hourly gust speeds and generate hourly power estimations
- Wind analysis needs to consider ideal locations for turbine installations at each zip code
- Microgrid analysis needs to include a method for analysis of space constraints
- Energy resource recommendations would ideally consider how diversified resources would more reliably power a grid, notably for data seasonality

